Final project R analysis: vitamin D, preeclampsia, and gene expression

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```
library(ggplot2)
library(knitr)
library(kableExtra)
library(pheatmap)
```

Purpose

I continued my Submission 2 analysis of CYP24A1, UTS2, and RUNDC3A, with baseline vitamin D as the continuous covariate and pregnancy outcome and vitamin D group (derived using 30 ng/mL) as the categorical covariates. I added maternal age and BMI for the summary table. The separate LaTeX report presents my findings.

Import and match the original data

```
# Read  files
covar_raw <- read.csv("GSE85307_covar-1.csv")
expr_raw <- read.csv("GSE85307_geneExpression.csv", row.names = 1)
link_raw <- read.csv("GSE85307_geneNames.csv")
```

```
# Keep the raw data and clean a working copy.
clean <- covar_raw
for (v in names(clean)[3:10]) {
  # Remove everything up to the last ": " so only the value remains.
  clean[, v] <- sub(".*: ", "", clean[, v])
}
```

```
# Convert the five continuous covariates from text to numbers.
num_vars <- c("mat_age_yrs", "delivery_gestAge_wks",
              "sampleCollection_gestAge_wks", "mat_BMI", "mat_vitD")
for (v in num_vars) {
  clean[, v] <- as.numeric(clean[, v])
}
```

```r
# Check unique sample IDs and membership before arranging columns.
length(table(clean$geoID)) == nrow(clean)
```

```
## [1] TRUE
```

```r
length(table(colnames(expr_raw))) == ncol(expr_raw)
```

```
## [1] TRUE
```

```r
all(clean$geoID %in% colnames(expr_raw))
```

```
## [1] TRUE
```

```r
all(colnames(expr_raw) %in% clean$geoID)
```

```
## [1] TRUE
```

```r
# Select expression columns by sample ID, not by their original positions.
expr <- expr_raw[, clean$geoID]
all(colnames(expr) == clean$geoID)
```

```
## [1] TRUE
```

```r
# Match probe IDs to gene names using named rows in the linker file.
rownames(link_raw) <- as.character(link_raw$PROBEID)
all(rownames(expr) %in% rownames(link_raw))
```

```
## [1] TRUE
```

```r
gene_names <- link_raw[rownames(expr), "GENE"]
sum(is.na(gene_names))
```

```
## [1] 0
```

```r
length(table(gene_names)) == nrow(expr)
```

```
## [1] TRUE
```

```r
rownames(expr) <- gene_names
```

```r
# Define the outcome factor and the derived vitamin D group.
clean$preg_preeclampsia <- factor(clean$preg_preeclampsia,
                                  levels = c("Normal", "Preeclampsia"))
clean$vitD_group <- factor(
  ifelse(clean$mat_vitD < 30, "<30 ng/mL", ">=30 ng/mL"),
  levels = c("<30 ng/mL", ">=30 ng/mL"))
```

```r
# Transpose to one participant per row, then combine the matched data.
all_genes <- as.data.frame(t(expr))
analysis_data <- cbind(clean, all_genes)
```

```r
# Inspect dimensions, missing values, and the four subgroup counts.
# No participants were removed, so the final sample size is 157.
dim(expr)
```

```
## [1] 500 157
```

```r
nrow(analysis_data)
```

```
## [1] 157
```

```r
sum(is.na(clean))
```

```
## [1] 0
```

```r
sum(is.na(expr))
```

```
## [1] 0
```

```r
table(clean$preg_preeclampsia, clean$vitD_group)
```

```
##
##                <30 ng/mL >=30 ng/mL
##   Normal              78         32
##   Preeclampsia        41          6
```

Plotting function from Submission 2

```r
# Produce the three plot types for one or more genes.
# categorical contains the names of exactly two categorical covariates.
# continuous_label and categorical_labels set readable axis and legend titles.

make_gene_plots <- function(data, genes, continuous, categorical,
                            continuous_label = continuous,
                            categorical_labels = categorical) {
  plots <- list()

  # Repeat the same plotting steps for every requested gene.
  for (gene in genes) {
    # Temporary names allow the function to accept column names.
    plot_data <- data.frame(
      Expression = data[, gene],
      Continuous = data[, continuous],
      Cat1 = data[, categorical[1]],
```

```
      Cat2 = data[, categorical[2]])

    # Plot 1: histogram
    histogram <- ggplot(plot_data, aes(x = Expression)) +
      geom_histogram(bins = 24, fill = "#337C8E", colour = "white") +
      labs(title = paste(gene, "expression distribution"),
           x = paste(gene, "expression (provided scale)"),
           y = "Number of participants") +
      theme_classic(base_size = 11)

    # Plot 2: scatterplot
    scatterplot <- ggplot(plot_data,
                          aes(x = Continuous, y = Expression)) +
      geom_point(colour = "#337C8E", alpha = 0.65, size = 1.7) +
      geom_smooth(method = "lm", formula = y ~ x,
                  colour = "#B65E27", fill = "#E4B48B") +
      labs(title = paste(gene, "expression vs.", continuous_label),
           x = continuous_label,
           y = paste(gene, "expression (provided scale)")) +
      theme_classic(base_size = 11)

    # Plot 3: boxplot
    boxplot <- ggplot(plot_data,
                      aes(x = Cat1, y = Expression, fill = Cat2)) +
      geom_boxplot(width = 0.6, position = position_dodge(0.75)) +
      scale_fill_manual(values = c("#337C8E", "#D5A13B")) +
      labs(title = paste(gene, "expression by group"),
           x = categorical_labels[1], fill = categorical_labels[2],
           y = paste(gene, "expression (provided scale)")) +
      theme_classic(base_size = 11) +
      theme(legend.position = "top")

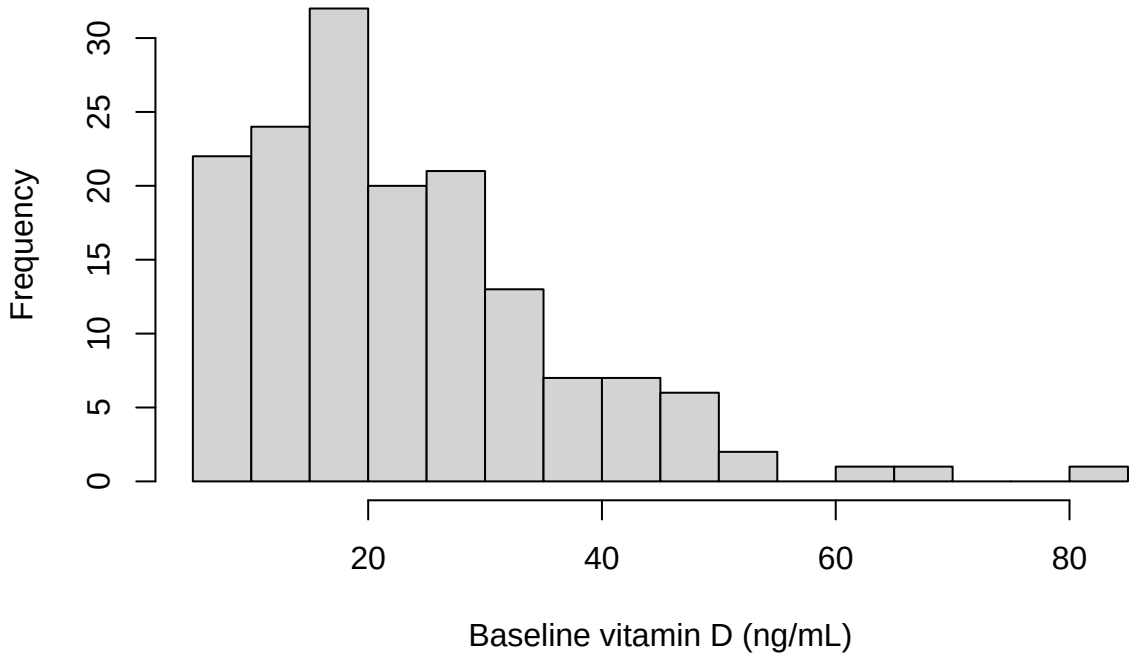
    # Save plots in a named list so they can be displayed separately.
    plots[[gene]] <- list(histogram = histogram,
                          scatterplot = scatterplot,
                          boxplot = boxplot)
  }
  plots
}
```

```
# The same three genes as Submission 2, with readable labels for the plots.
plots <- make_gene_plots(
  data = analysis_data,
  genes = c("CYP24A1", "UTS2", "RUNDC3A"),
  continuous = "mat_vitD",
  categorical = c("preg_preeclampsia", "vitD_group"),
  continuous_label = "Baseline 25(OH)D (ng/mL)",
  categorical_labels = c("Pregnancy outcome", "Baseline vitamin D"))
```

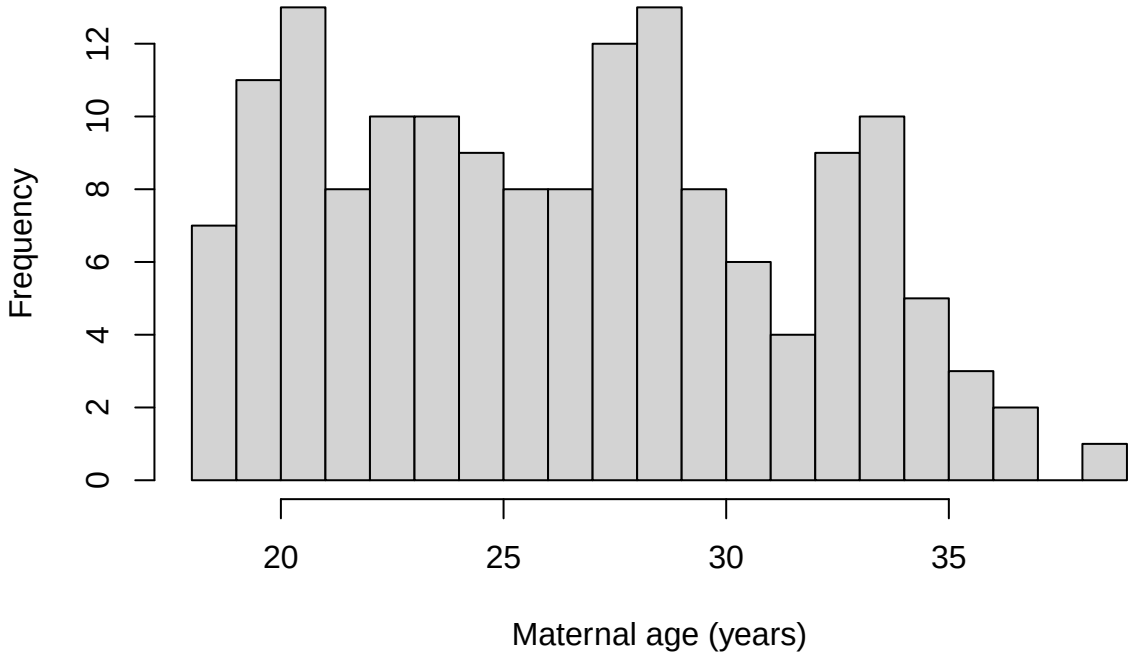
Check continuous covariate distributions

I inspected the three continuous covariates before choosing summary statistics. Age looked roughly symmetric, so I used mean (SD). Vitamin D and BMI were right-skewed, so I used median [Q1, Q3].

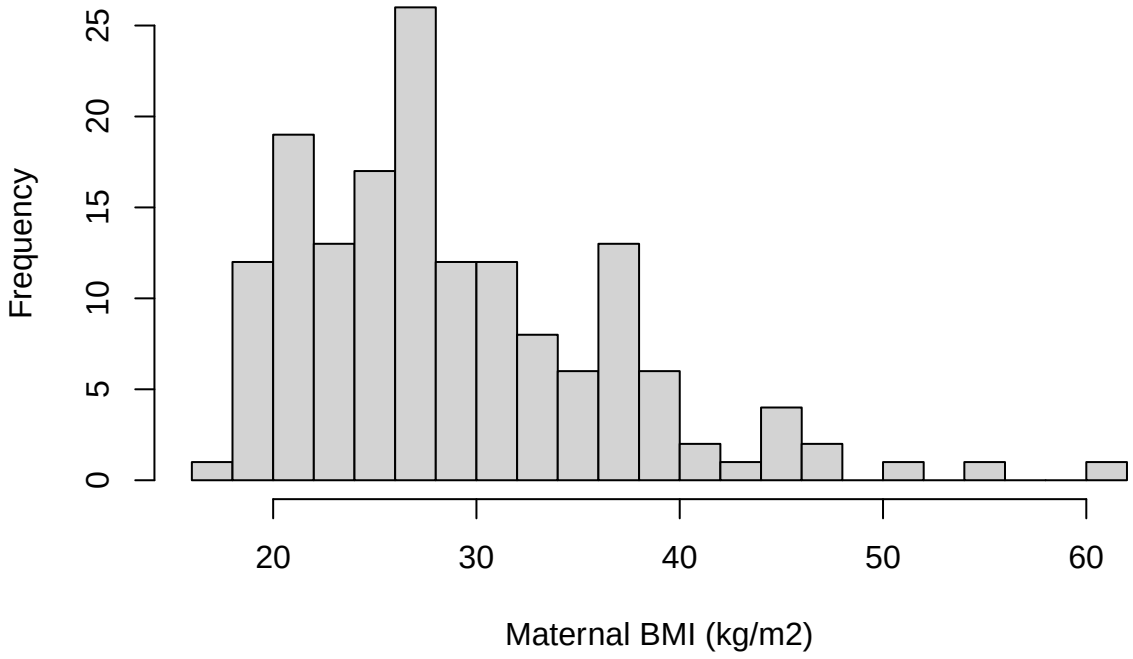
```
# Distribution of baseline vitamin D.
hist(clean$mat_vitD, breaks = 20, main = "",
     xlab = "Baseline vitamin D (ng/mL)")
```




```
# Distribution of maternal age.
hist(clean$mat_age_yrs, breaks = 20, main = "",
     xlab = "Maternal age (years)")
```




```
# Distribution of maternal BMI.
hist(clean$mat_BMI, breaks = 20, main = "",
     xlab = "Maternal BMI (kg/m2)")
```



Summary statistics table

```
# Step 1: Summary functions from Lecture 09.
meanSD <- function(x) {
  paste0(round(mean(x), 2), " (", round(sd(x), 2), ")")
}

medianIQR <- function(x) {
  paste0(round(median(x), 2), " [",
         round(quantile(x, 1/4), 2), ", ",
         round(quantile(x, 3/4), 2), "]")
}

# n (%) for every level of a categorical variable.
catNP <- function(x) {
  paste0(table(x), " (", round(table(x) / length(x) * 100, 1), ")")
}
```

```
# Step 2: Stratify the table by pregnancy outcome.
normal <- clean[clean$preg_preeclampsia == "Normal", ]
preeclampsia <- clean[clean$preg_preeclampsia == "Preeclampsia", ]
```

```r
# Step 3: Three continuous and two categorical covariates.
table1 <- data.frame(
  Variable = c("Vitamin D (ng/mL), median [Q1, Q3]",
               "Maternal age (years), mean (SD)",
               "Maternal BMI (kg/m2), median [Q1, Q3]",
               "Pregnancy outcome, n (%)", "Normal", "Preeclampsia",
               "Vitamin D group, n (%)", "<30 ng/mL", ">=30 ng/mL"),
  Overall = c(medianIQR(clean$mat_vitD), meanSD(clean$mat_age_yrs),
              medianIQR(clean$mat_BMI),
              "", catNP(clean$preg_preeclampsia),
              "", catNP(clean$vitD_group)),
  Normal = c(medianIQR(normal$mat_vitD), meanSD(normal$mat_age_yrs),
             medianIQR(normal$mat_BMI),
             "", catNP(normal$preg_preeclampsia),
             "", catNP(normal$vitD_group)),
  Preeclampsia = c(medianIQR(preeclampsia$mat_vitD),
                   meanSD(preeclampsia$mat_age_yrs),
                   medianIQR(preeclampsia$mat_BMI),
                   "", catNP(preeclampsia$preg_preeclampsia),
                   "", catNP(preeclampsia$vitD_group)))
```

```r
# Step 4: Column names show the size of each group.
table_names <- c("Variable",
                 paste0("Overall (n = ", nrow(clean), ")"),
                 paste0("Normal (n = ", nrow(normal), ")"),
                 paste0("Preeclampsia (n = ", nrow(preeclampsia), ")"))
```

```r
# Step 5: Print the formatted table, following Lecture 09.
kable(table1, caption = "Participant characteristics by pregnancy outcome.",
      col.names = table_names, align = c("l", "r", "r", "r")) %>%
  add_indent(positions = c(5, 6, 8, 9)) %>%
  kable_classic()
```

Table 1: Participant characteristics by pregnancy outcome.

Variable	Overall (n = 157)	Normal (n = 110)	Preeclampsia (n = 47)
Vitamin D (ng/mL), median [Q1, Q3]	20.4 [13.4, 29.5]	21.8 [13.85, 32.08]	17.7 [12.55, 27.4]
Maternal age (years), mean (SD)	26.38 (5.04)	26.63 (5.08)	25.78 (4.95)
Maternal BMI (kg/m2), median [Q1, Q3]	27.33 [22.9, 33.27]	26.54 [22.35, 30.97]	31.24 [24.62, 36.41]
Pregnancy outcome, n (%)			
Normal	110 (70.1)	110 (100)	0 (0)
Preeclampsia	47 (29.9)	0 (0)	47 (100)
Vitamin D group, n (%)			
<30 ng/mL	119 (75.8)	78 (70.9)	41 (87.2)
>=30 ng/mL	38 (24.2)	32 (29.1)	6 (12.8)

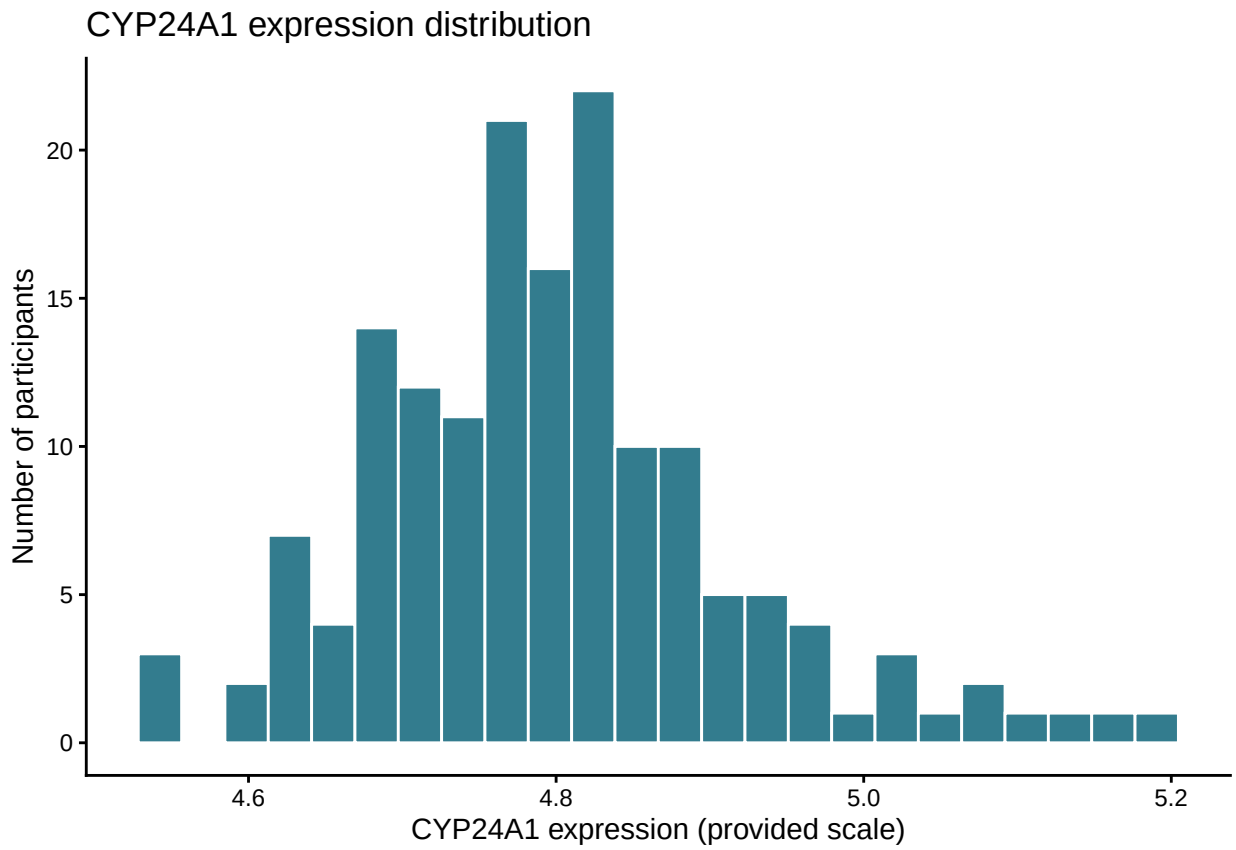
```r
# Step 6: Save the same table as LaTeX code for the LaTeX report.
table_latex <- kable(table1, format = "latex",
                     caption = "Participant characteristics by pregnancy outcome.",
                     col.names = table_names, align = c("l", "r", "r", "r")) %>%
  add_indent(positions = c(5, 6, 8, 9)) %>%
```

```
  kable_classic()
writeLines(table_latex, "Table1.tex")
```

I found lower median vitamin D and higher median BMI in the preeclampsia group. Mean age was similar between outcome groups (Table 1).

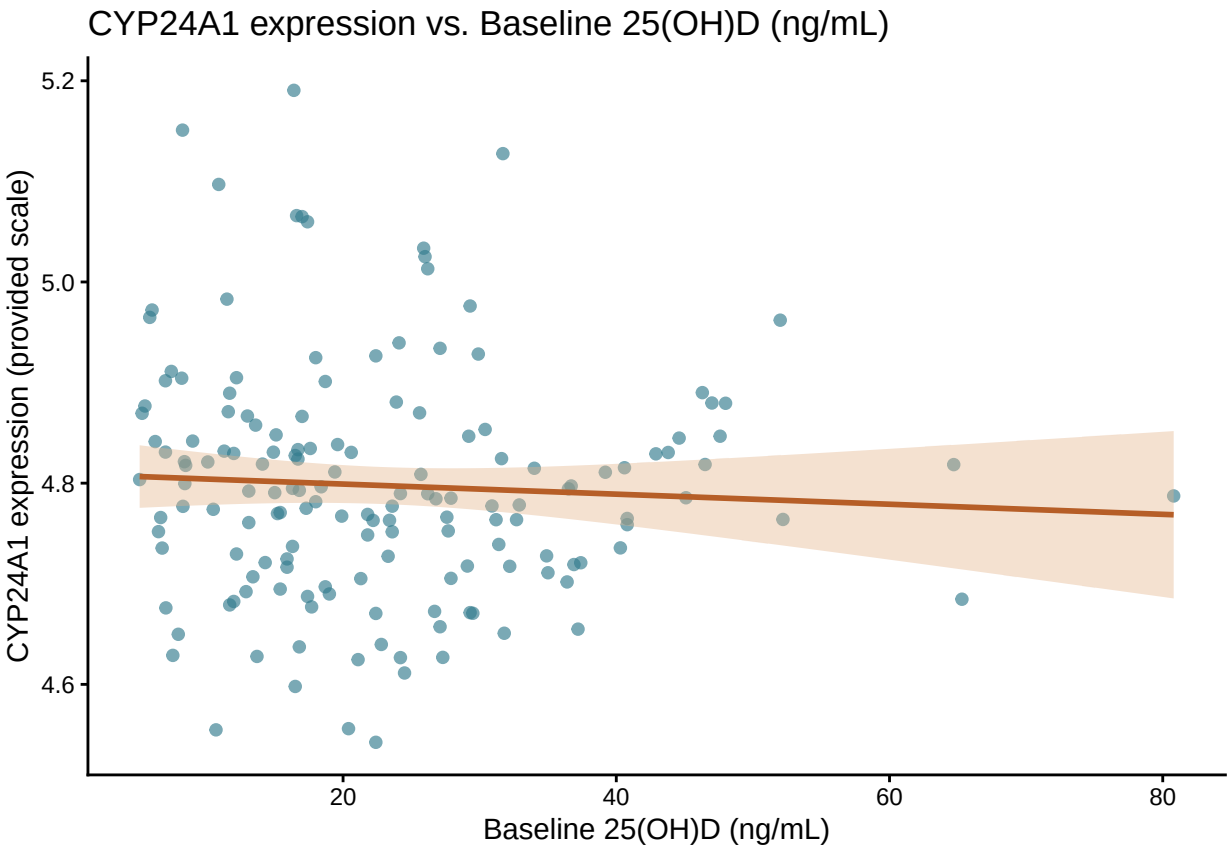
CYP24A1 plots

```
# Figure 1: CYP24A1 expression histogram.
plots$CYP24A1$histogram
```



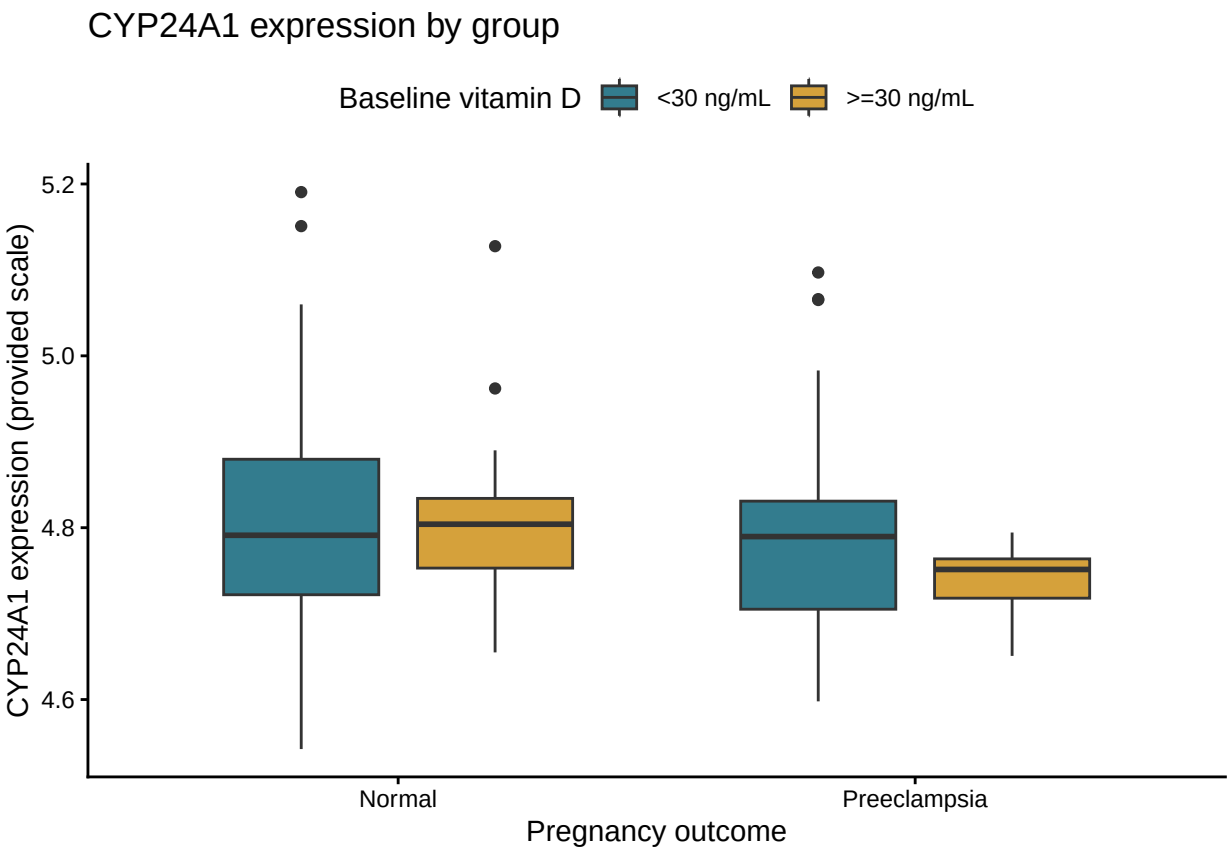
I observed a single main peak near 4.8, with a small upper tail.

```
# Figure 2: CYP24A1 vs. baseline vitamin D.
plots$CYP24A1$scatterplot
```



I observed a nearly flat fitted line. Vitamin D did not show a clear linear pattern with CYP24A1 expression in this sample.

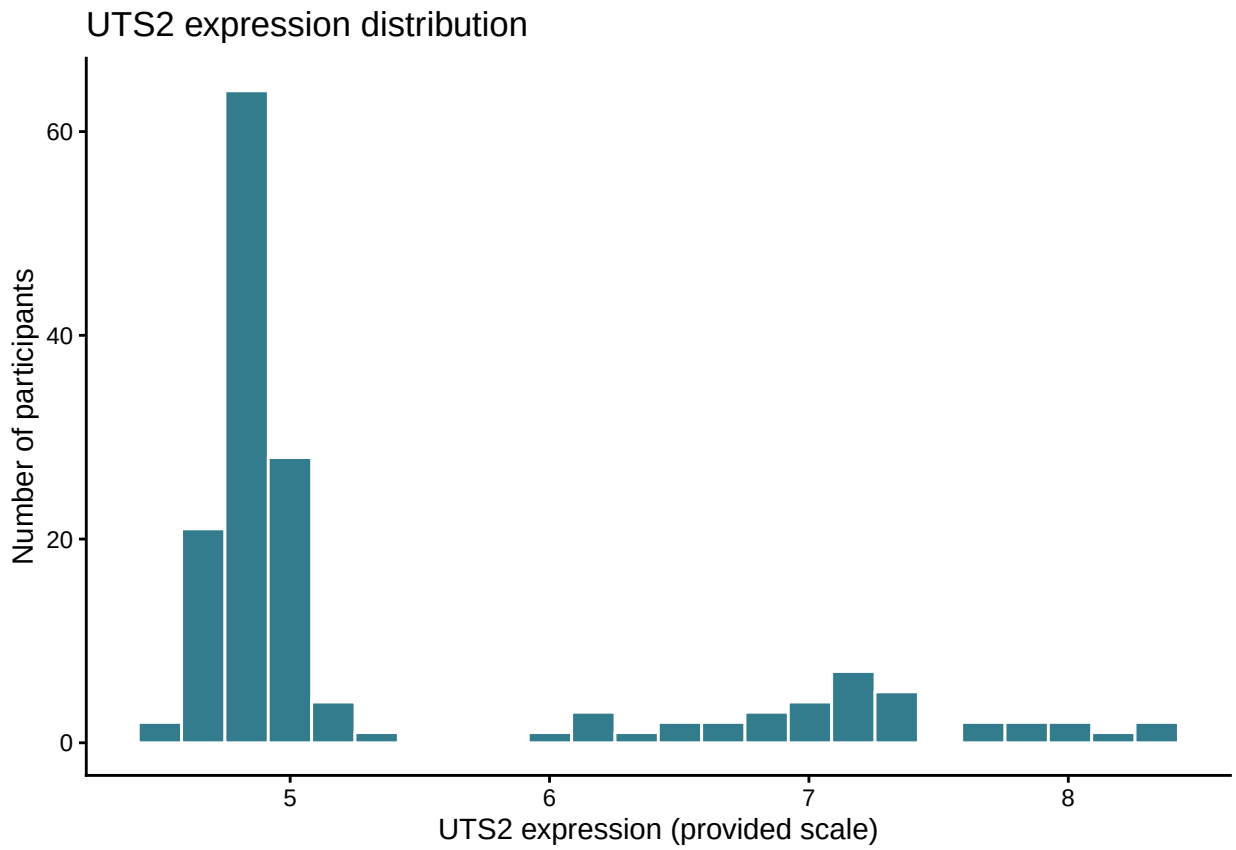
```
# Figure 3: CYP24A1 by outcome and vitamin D group.
plots$CYP24A1$boxplot
```



I observed overlapping expression distributions. The subgroup with preeclampsia and vitamin D at least 30 ng/mL had only six participants, so I interpreted its lower median cautiously.

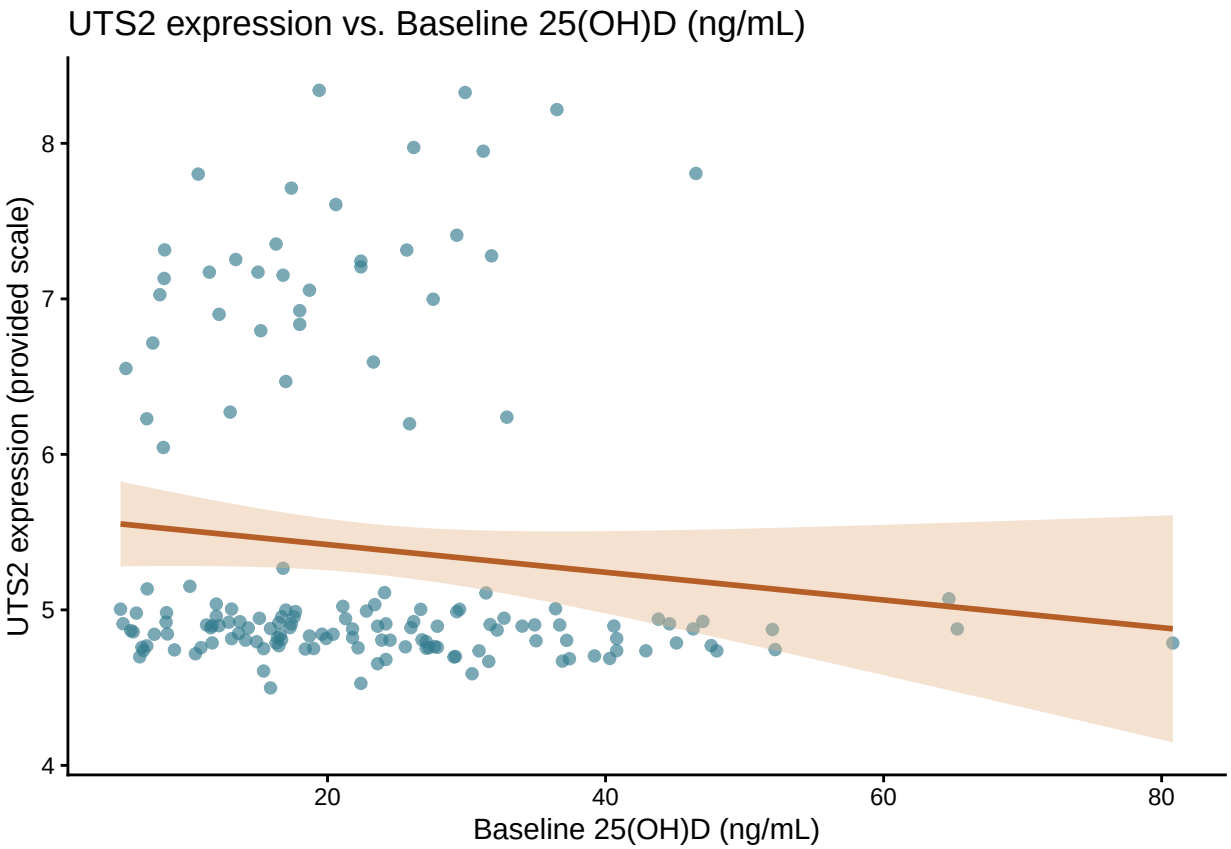
UTS2 plots

```
# Figure 4: UTS2 expression histogram.
plots$UTS2$histogram
```



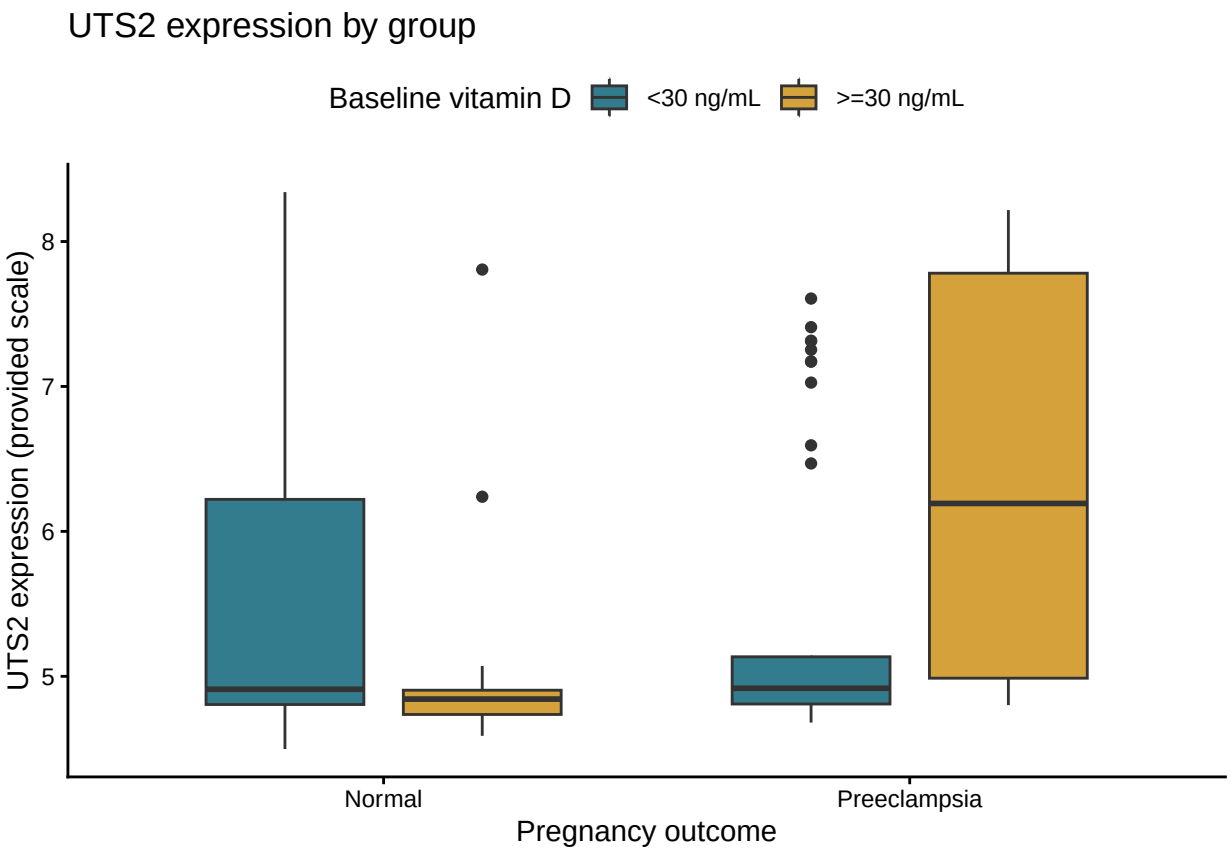
I observed two groups of UTS2 values: most participants were near 5, and a smaller group was between about 6 and 8.5.

```
# Figure 5: UTS2 vs. baseline vitamin D.
plots$UTS2$scatterplot
```



Both UTS2 groups appeared across the range of vitamin D, so the single fitted line passed between them and did not describe either group well.

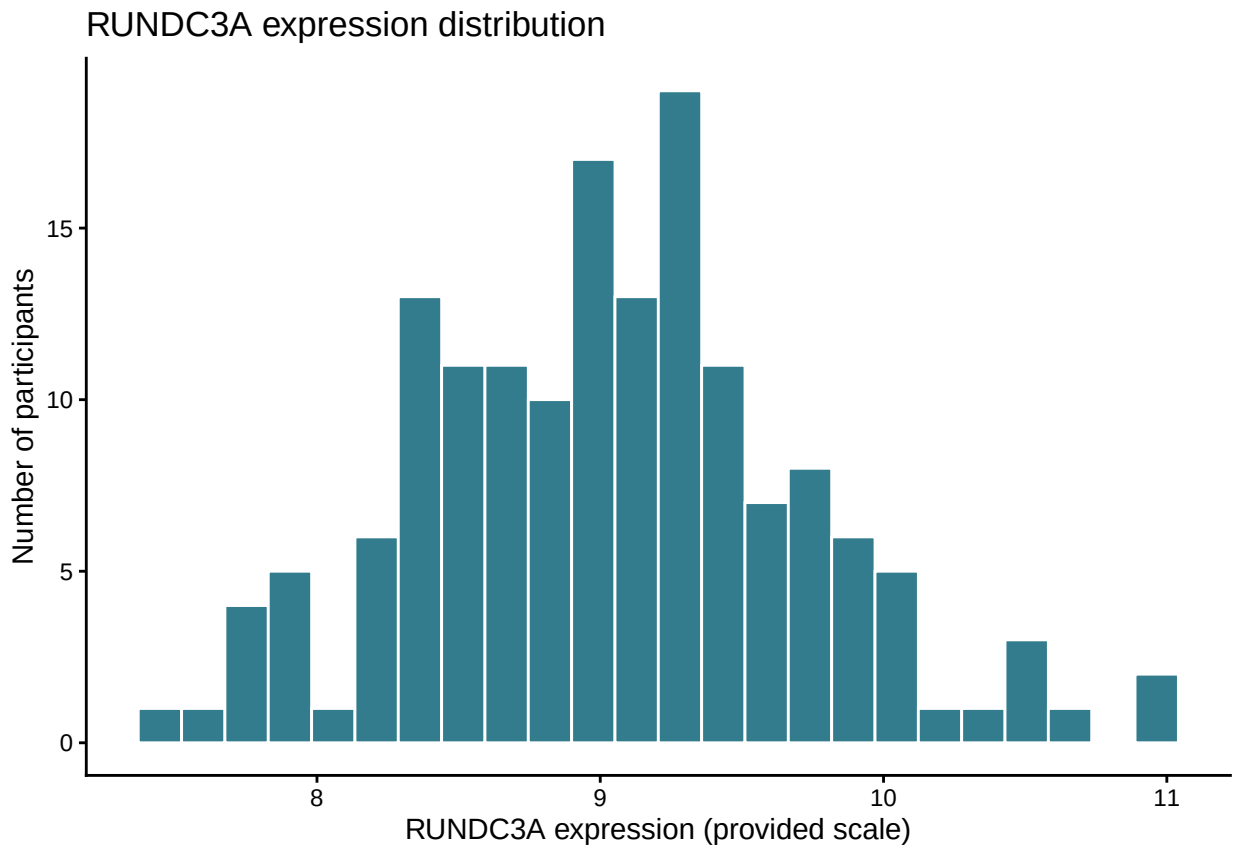
```
# Figure 6: UTS2 by outcome and vitamin D group.
plots$UTS2$boxplot
```



I observed a higher median UTS2 expression among the six preeclampsia participants in the higher vitamin D group. I did not treat this small subgroup comparison as evidence of a difference.

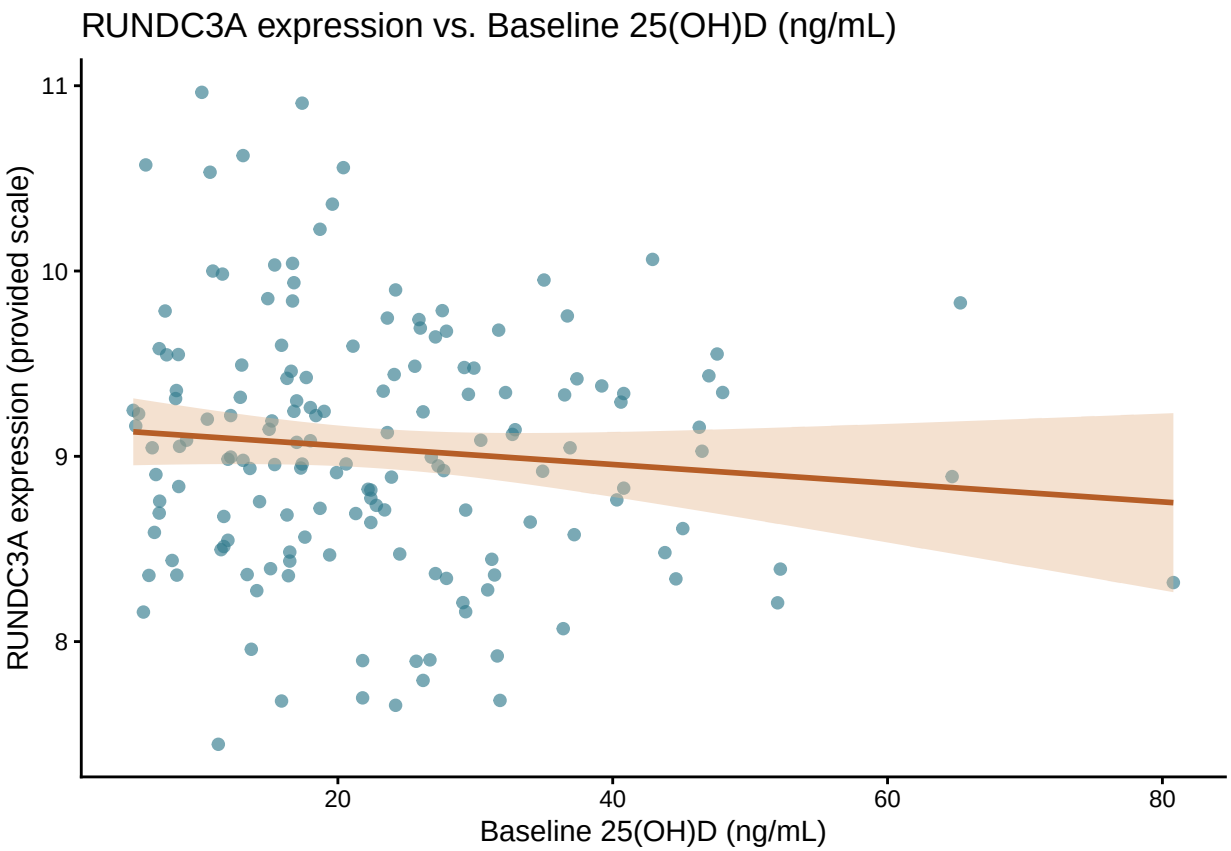
RUNDC3A plots

```
# Figure 7: RUNDC3A expression histogram.
plots$RUNDC3A$histogram
```



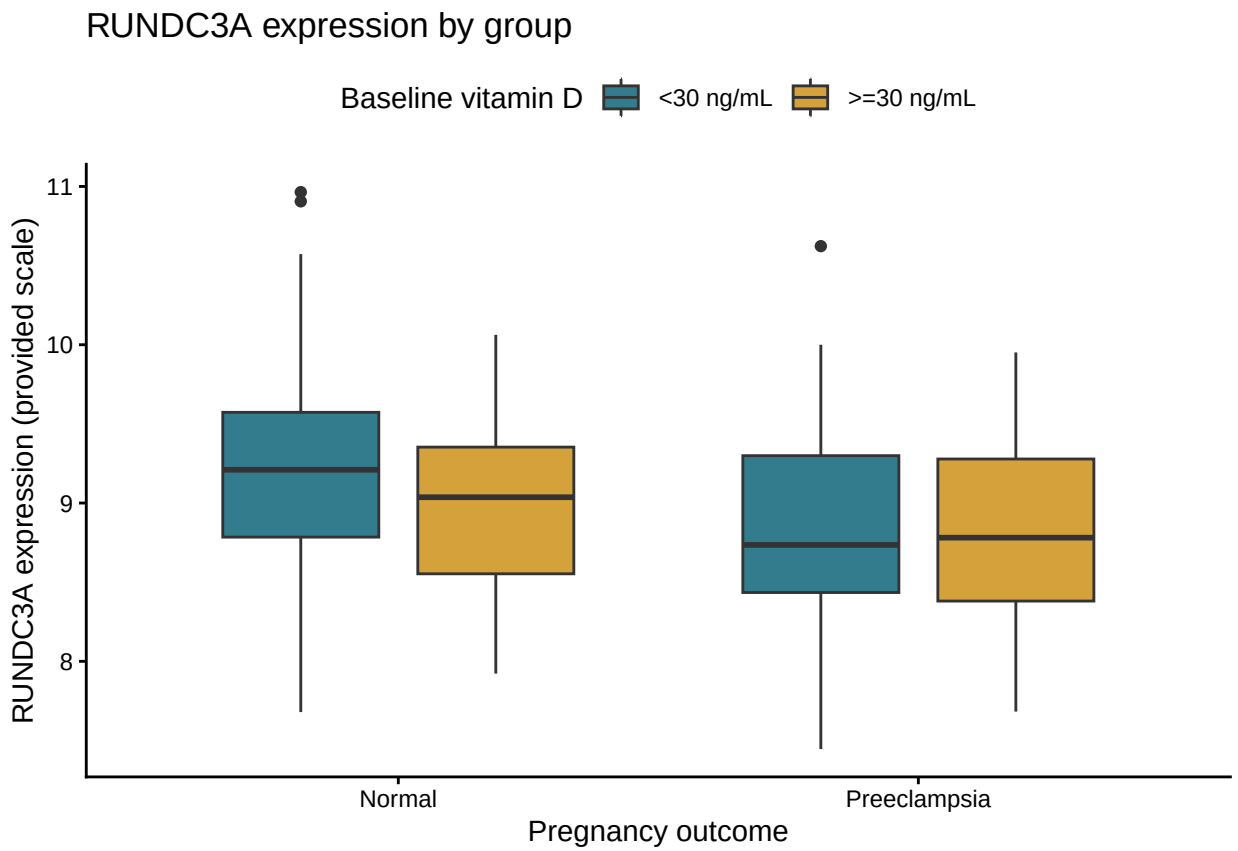
I observed a broader distribution centered near 9 on the supplied scale.

```
# Figure 8: RUNDC3A vs. baseline vitamin D.
plots$RUNDC3A$scatterplot
```



I observed a slight downward fitted line, with substantial scatter around it.

```
# Figure 9: RUNDC3A by outcome and vitamin D group.
plots$RUNDC3A$boxplot
```



I observed lower median RUNDC3A expression in the preeclampsia group within both vitamin D groups. The distributions still overlapped, and I did not perform a hypothesis test.

Ten-gene clustered heatmap (Figure 10)

I selected the ten genes with the highest standard deviation across the 500 supplied genes, following Lecture 08. I kept the supplied expression scale and did not standardize rows.

```
# Figure 10: clustered heatmap of the ten most variable genes.

# 1. Select the ten genes with the highest standard deviation.
# Ranking by SD gives the same genes as ranking by variance.
gene_sd <- apply(expr, MARGIN = 1, FUN = sd)
heatmap_data <- expr[order(gene_sd, decreasing = TRUE)[1:10], ]

# 2. Tracking bar data, with one row per sample column of the heatmap.
# check.names = FALSE keeps the spaces in the tracking bar labels.
annotation_data <- data.frame(
  row.names = clean$geoID,
  "Pregnancy outcome" = clean$preg_preeclampsia,
  "Vitamin D" = clean$vitD_group,
  check.names = FALSE)

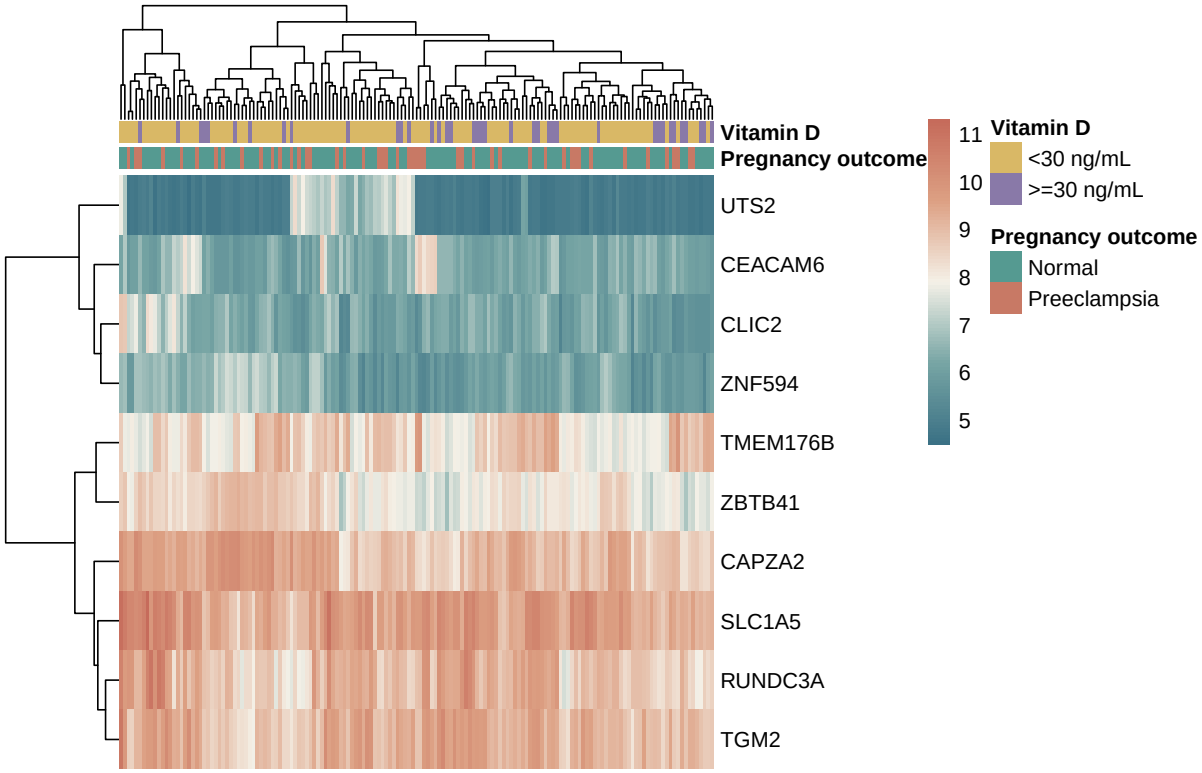
# Check that the sample IDs match the heatmap columns.
all(rownames(annotation_data) == colnames(heatmap_data))
```

```
## [1] TRUE
```

```
# 3. Different colors for each tracking bar
# The list names must match the tracking bar labels above.
annotation_colors <- list(
  "Pregnancy outcome" = c("Normal" = "#559A91", "Preeclampsia" = "#C87965"),
  "Vitamin D" = c("<30 ng/mL" = "#D9B866", ">=30 ng/mL" = "#8979A8"))

# 4. Color gradient for gene expression, from low (blue) to high (red).
heatmap_colors <- colorRampPalette(
  c("#3B7185", "#78A6A5", "#F5F1E8", "#DDA184", "#C66B5B"))(100)

# 5. Clustered heatmap
pheatmap(heatmap_data,
         color = heatmap_colors,
         clustering_distance_cols = "euclidean",
         clustering_distance_rows = "euclidean",
         annotation_col = annotation_data,
         annotation_colors = annotation_colors,
         show_colnames = FALSE,
         border_color = NA)
```



The genes separated mainly by overall expression level, and one cluster of participants had higher UTS2 expression. The tracking bars did not separate by outcome or vitamin D group. I treated the clustering as exploratory and did not use the dendrogram as a test of group differences.

Additional plot: dumbbell plot (Figure 11)

I used a dumbbell plot to compare the median expression of the ten heatmap genes between the two outcome groups. Each line joins the two group medians for one gene, and genes are ordered by the difference.

```
# Figure 11: median expression of the ten heatmap genes by outcome.

# Sample IDs for each outcome group.
normal_ids <- clean$geoID[clean$preg_preeclampsia == "Normal"]
pe_ids <- clean$geoID[clean$preg_preeclampsia == "Preeclampsia"]

# Median expression of each heatmap gene within each outcome group.
dumbbell_data <- data.frame(
  Gene = rownames(heatmap_data),
  Normal = apply(heatmap_data[, normal_ids], MARGIN = 1, FUN = median),
  Preeclampsia = apply(heatmap_data[, pe_ids], MARGIN = 1, FUN = median))

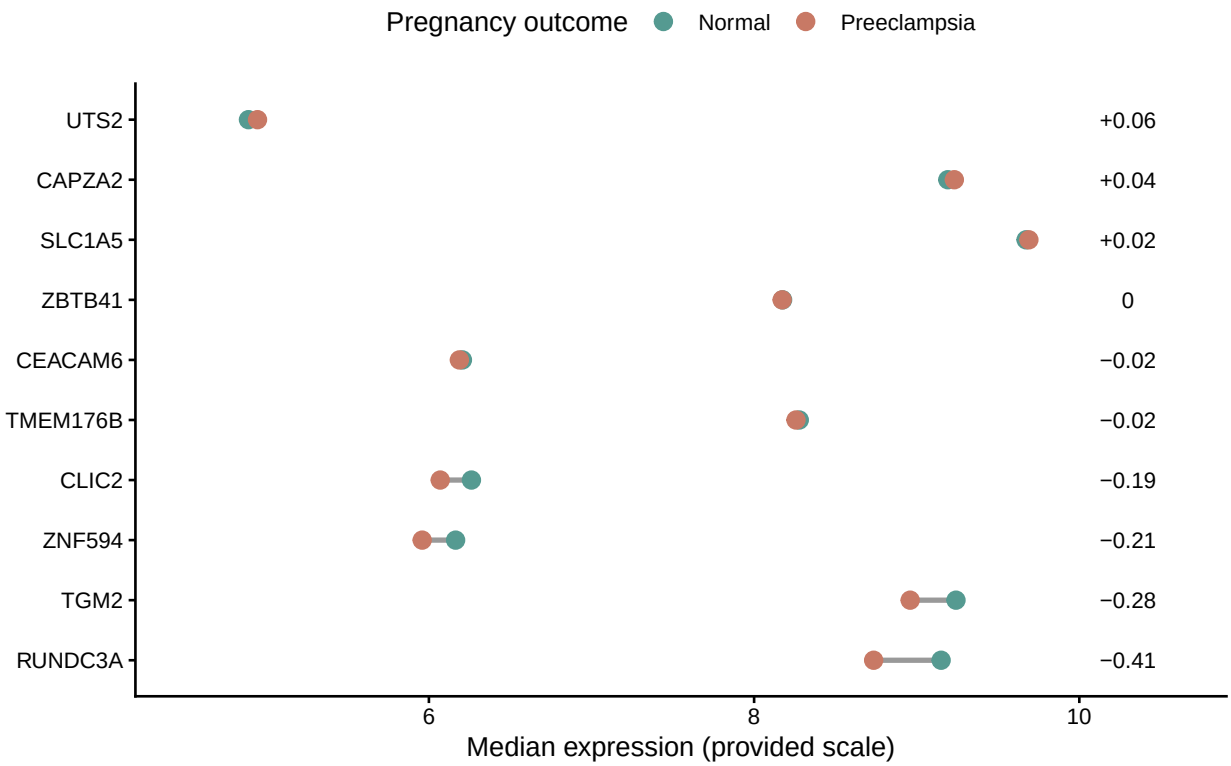
# Difference between groups (Preeclampsia minus Normal).
dumbbell_data$Difference <- dumbbell_data$Preeclampsia - dumbbell_data$Normal

# Order genes from the largest decrease to the largest increase.
dumbbell_data$Gene <- factor(dumbbell_data$Gene,
  levels = dumbbell_data$Gene[order(dumbbell_data$Difference)])

# One row per gene and outcome, for the colored points.
point_data <- data.frame(
  Gene = rep(dumbbell_data$Gene, 2),
  Outcome = rep(c("Normal", "Preeclampsia"), each = nrow(dumbbell_data)),
  Median = c(dumbbell_data$Normal, dumbbell_data$Preeclampsia))

# Difference labels, with a plus sign for increases.
diff_label <- round(dumbbell_data$Difference, 2)
diff_label <- paste0(ifelse(diff_label > 0, "+", ""), diff_label)

# Segments join the two group medians; points show each group.
# Outcome colors match the heatmap tracking bar.
ggplot() +
  geom_segment(data = dumbbell_data,
               aes(x = Normal, xend = Preeclampsia, y = Gene, yend = Gene),
               colour = "grey60", linewidth = 1) +
  geom_point(data = point_data,
             aes(x = Median, y = Gene, colour = Outcome), size = 3) +
  annotate("text", x = 10.3, y = as.character(dumbbell_data$Gene),
           label = diff_label, size = 3.2) +
  scale_colour_manual(values = c("Normal" = "#559A91",
                                 "Preeclampsia" = "#C87965")) +
  xlim(4.5, 10.6) +
  labs(x = "Median expression (provided scale)", y = NULL,
       colour = "Pregnancy outcome") +
  theme_classic(base_size = 11) +
  theme(legend.position = "top")
```



RUNDC3A showed the largest difference, with lower median expression in preeclampsia (8.74 vs. 9.15). TGM2, ZNF594, and CLIC2 were also about 0.2 to 0.3 lower. The remaining six genes differed by 0.06 or less.

Software versions

```
# Versions to cite in the Methods section of the LaTeX report.
R.version.string
```

```
## [1] "R version 4.6.1 (2026-06-24 ucrt)"
```

```
packageVersion("ggplot2")
```

```
## [1] '4.0.3'
```

```
packageVersion("knitr")
```

```
## [1] '1.52'
```

```
packageVersion("kableExtra")
```

```
## [1] '1.4.1'
```

```r
packageVersion("pheatmap")
```

```
## [1] '1.0.13'
```

Note on resources used

This was my first time using R, so I made frequent errors while writing the code. I used an AI assistant (Claude) to help me debug errors and understand error messages, and I watched YouTube tutorials to help me understand R and ggplot2 concepts. I reviewed and ran all of the code myself, and I checked the results against the data.